

maintenance_analyse

August 15, 2026

1 1.Data Analyse

This section analyzes the machine operating data collected during August 2026.

The objective is to evaluate the main machine parameters and identify abnormal operating conditions that may be associated with machine failures.

The following parameters are analyzed:

- **Temperature (°C)**
- **Rotational speed (rpm)**
- **Torque (Nm)**
- **Machine failure status**

The analysis uses: - **Pandas** for data loading, filtering, and manipulation. - **NumPy** for numerical calculations. - **Matplotlib** for data visualization.

```
[2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

1.1 1.1 Loading the Dataset

The machine data is stored in a CSV file. The dataset is loaded into a Pandas DataFrame to facilitate data exploration and analysis.

```
[8]: df = pd.read_csv("Real_machine_data.csv")

print(df)
```

	Time	Temperature	Rotational speed	Torque	Failure
0	2026-08-01 09:00:00	35.5	1551	42.8	0
1	2026-08-01 10:00:00	35.6	1408	46.3	0
2	2026-08-01 11:00:00	35.4	1498	49.4	0
3	2026-08-01 12:00:00	35.5	1433	39.5	0
4	2026-08-01 13:00:00	35.6	1408	40.0	0
...	...	...	...	...	...
1219	2026-12-31 12:00:00	35.0	1357	41.8	0
1220	2026-12-31 13:00:00	35.1	1577	29.4	0
1221	2026-12-31 14:00:00	35.2	1399	46.4	0
1222	2026-12-31 15:00:00	35.2	1399	59.0	0

```
1223  2026-12-31 16:00:00          35.2          1492          48.1          0
```

```
[1224 rows x 5 columns]
```

1.2 1.2 Data Preparation and Time Filtering

The `Time` column is converted to datetime format to allow chronological filtering.

For this analysis, only measurements collected during **August 2026** are considered.

```
[9]: df["Time"] = pd.to_datetime(df["Time"])

mask = (
    (df["Time"] >= "2026-08-01") &
    (df["Time"] < "2026-09-01")
)

Time = df.loc[mask, "Time"]
temperature = df.loc[mask, "Temperature"]
R_speed = df.loc[mask, "Rotational speed"]
torque = df.loc[mask, "Torque"]
Failure = df.loc[mask, "Failure"]
```

1.3 1.3 Statistical Analysis of Machine Parameters

The main statistical indicators are calculated for temperature, rotational speed, and torque.

For each parameter, the following values are determined:

- Mean value
- Maximum value
- Minimum value

These indicators provide an overview of the machine's operating conditions during the analyzed period.

```
[10]: mean_temp = round(np.mean(temperature), 1)
mean_speed = round(np.mean(R_speed), 1)
mean_torque = round(np.mean(torque), 1)

max_temp = np.max(temperature)
max_speed = np.max(R_speed)
max_torque = np.max(torque)

min_temp = np.min(temperature)
min_speed = np.min(R_speed)
min_torque = np.min(torque)
```

```
[11]: print("==== PRODUCTION REPORT =====")
```

```

print("\nTemperature")
print("Average:", mean_temp, "°C")
print("Maximum:", max_temp, "°C")
print("Minimum:", min_temp, "°C")

print("\nRotation speed")
print("Average:", mean_speed, "rpm")
print("Maximum:", max_speed, "rpm")
print("Minimum:", min_speed, "rpm")

print("\nTorque")
print("Average:", mean_torque, "Nm")
print("Maximum:", max_torque, "Nm")
print("Minimum:", min_torque, "Nm")

```

===== PRODUCTION REPORT =====

Temperature

Average: 35.6 °C

Maximum: 36.4 °C

Minimum: 34.9 °C

Rotation speed

Average: 1566.6 rpm

Maximum: 2861 rpm

Minimum: 1200 rpm

Torque

Average: 39.9 Nm

Maximum: 65.7 Nm

Minimum: 4.6 Nm

1.4 1.4 Determination of Operating Limits

To identify abnormal operating conditions, reference limits are determined from measurements corresponding to normal machine operation.

Only observations where `Failure = 0` are considered.

The maximum observed value under normal operation is used as the reference limit for:

- Temperature
- Rotational speed
- Torque

Values exceeding these limits are considered potential abnormal conditions.

```

[12]: normal_data = df.loc[
        mask & (df["Failure"] == 0)
    ]

```

```

limit_temp = normal_data["Temperature"].max()
limit_speed = normal_data["Rotational speed"].max()
limit_torque = normal_data["Torque"].max()

```

```

[13]: print("Normal operating limits:")
      print("Temperature:", limit_temp, "°C")
      print("Rotation speed:", limit_speed, "rpm")
      print("Torque:", limit_torque, "Nm")

```

```

Normal operating limits:
Temperature: 36.4 °C
Rotation speed: 2035 rpm
Torque: 60.5 Nm

```

1.5 1.5 Detection of Abnormal Operating Conditions

The previously calculated normal operating limits are used to detect measurements that exceed the expected operating range.

An alarm is generated when:

- Temperature > normal temperature limit
- Rotational speed > normal rotational speed limit
- Torque > normal torque limit

These alarms indicate potentially abnormal machine operating conditions.

```

[14]: high_temperature = df.loc[
      mask & (df["Temperature"] > limit_temp)
      ]

      high_speed = df.loc[
      mask & (df["Rotational speed"] > limit_speed)
      ]

      high_torque = df.loc[
      mask & (df["Torque"] > limit_torque)
      ]

```

```

[15]: print("==== TEMPERATURE ALARMS ====")

      for index, row in high_temperature.iterrows():
          print(
              row["Time"],
              "→",
              row["Temperature"],
              "°C"
          )

```

===== TEMPERATURE ALARMS =====

```
[16]: print("===== ROTATION SPEED ALARMS =====")

for index, row in high_speed.iterrows():
    print(
        row["Time"],
        "→",
        row["Rotational speed"],
        "rpm"
    )
```

===== ROTATION SPEED ALARMS =====

```
2026-08-02 09:00:00 → 2521 rpm
2026-08-07 10:00:00 → 2261 rpm
2026-08-07 11:00:00 → 2861 rpm
2026-08-12 11:00:00 → 2442 rpm
2026-08-14 10:00:00 → 2308 rpm
2026-08-18 16:00:00 → 2253 rpm
2026-08-22 11:00:00 → 2425 rpm
2026-08-23 14:00:00 → 2335 rpm
2026-08-24 15:00:00 → 2365 rpm
2026-08-25 11:00:00 → 2678 rpm
2026-08-27 09:00:00 → 2269 rpm
2026-08-29 11:00:00 → 2204 rpm
```

```
[17]: print("===== TORQUE ALARMS =====")

for index, row in high_torque.iterrows():
    print(
        row["Time"],
        "→",
        row["Torque"],
        "Nm"
    )
```

===== TORQUE ALARMS =====

```
2026-08-09 14:00:00 → 65.7 Nm
2026-08-10 10:00:00 → 63.0 Nm
2026-08-10 16:00:00 → 61.5 Nm
2026-08-13 09:00:00 → 65.3 Nm
2026-08-21 09:00:00 → 60.7 Nm
2026-08-22 09:00:00 → 62.3 Nm
2026-08-25 09:00:00 → 62.7 Nm
2026-08-26 16:00:00 → 60.7 Nm
2026-08-29 10:00:00 → 61.8 Nm
2026-08-31 13:00:00 → 63.9 Nm
```

1.6 1.6 Alarm Summary

The number of detected alarms is calculated for each machine parameter.

This provides an overview of which parameters exceeded their normal operating limits most frequently.

```
[18]: print("==== ALARM SUMMARY ====")

print("Temperature alarms:", len(high_temperature))
print("Rotation speed alarms:", len(high_speed))
print("Torque alarms:", len(high_torque))

total_alarms = (
    len(high_temperature)
    + len(high_speed)
    + len(high_torque)
)

print("Total alarms:", total_alarms)
```

```
==== ALARM SUMMARY ====
Temperature alarms: 0
Rotation speed alarms: 12
Torque alarms: 10
Total alarms: 22
```

1.7 1.7 Machine Failure Distribution

The machine observations are divided into two categories:

- Normal operation (Failure = 0)
- Machine failure (Failure = 1)

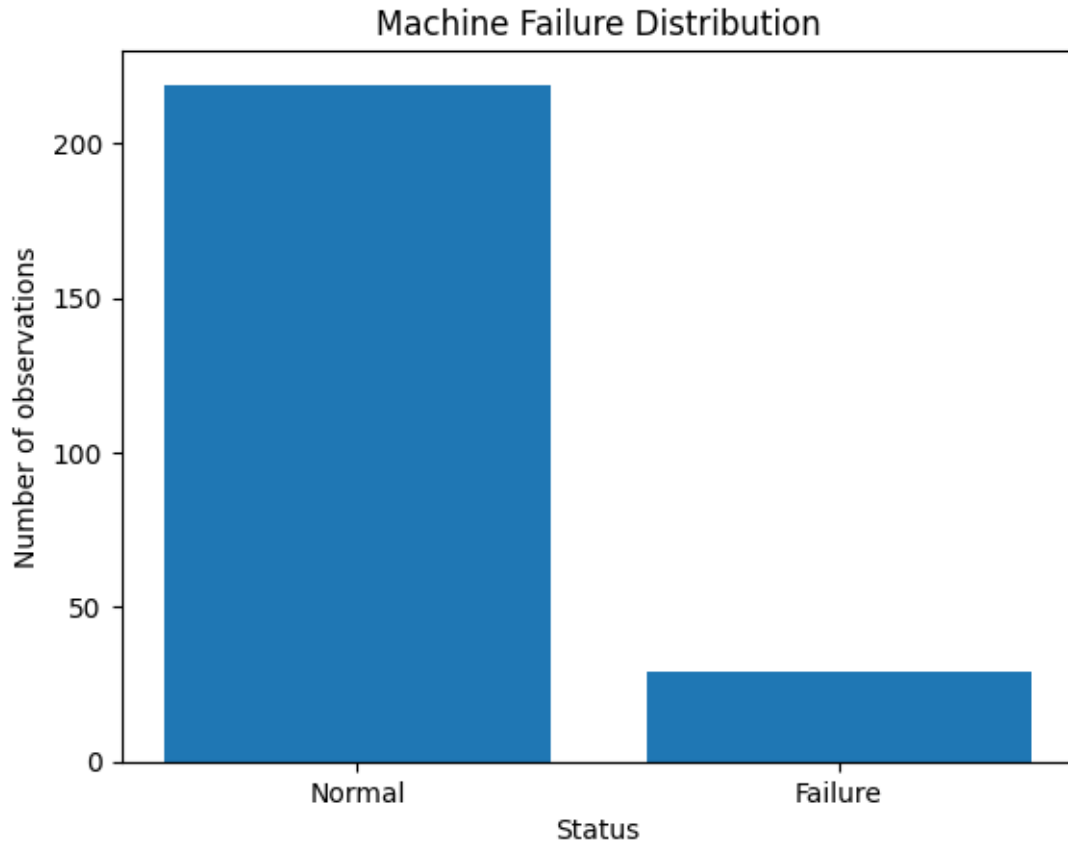
A bar chart is used to visualize the distribution of normal and failure states during August 2026.

```
[19]: failure = len(df.loc[mask & (df["Failure"] == 1)])
normal = len(df.loc[mask & (df["Failure"] == 0)])

plt.bar(
    ["Normal", "Failure"],
    [normal, failure]
)

plt.title("Machine Failure Distribution")
plt.xlabel("Status")
plt.ylabel("Number of observations")

plt.show()
```



1.8 Evolution of Machine Parameters

The evolution of temperature, rotational speed, and torque is visualized over time.

The reference operating limit calculated from normal machine operation is displayed on each graph.

This visualization makes it possible to identify measurements that exceed the normal operating range.

```
[20]: fig, axes = plt.subplots(3, 1, figsize=(12, 10))
```

```
# Temperature
axes[0].plot(Time, temperature)
axes[0].set_title("Temperature")
axes[0].axhline(
    limit_temp,
    linestyle="--",
    color="red",
    label="Normal operating limit"
)
axes[0].set_xlabel("Date")
```

```

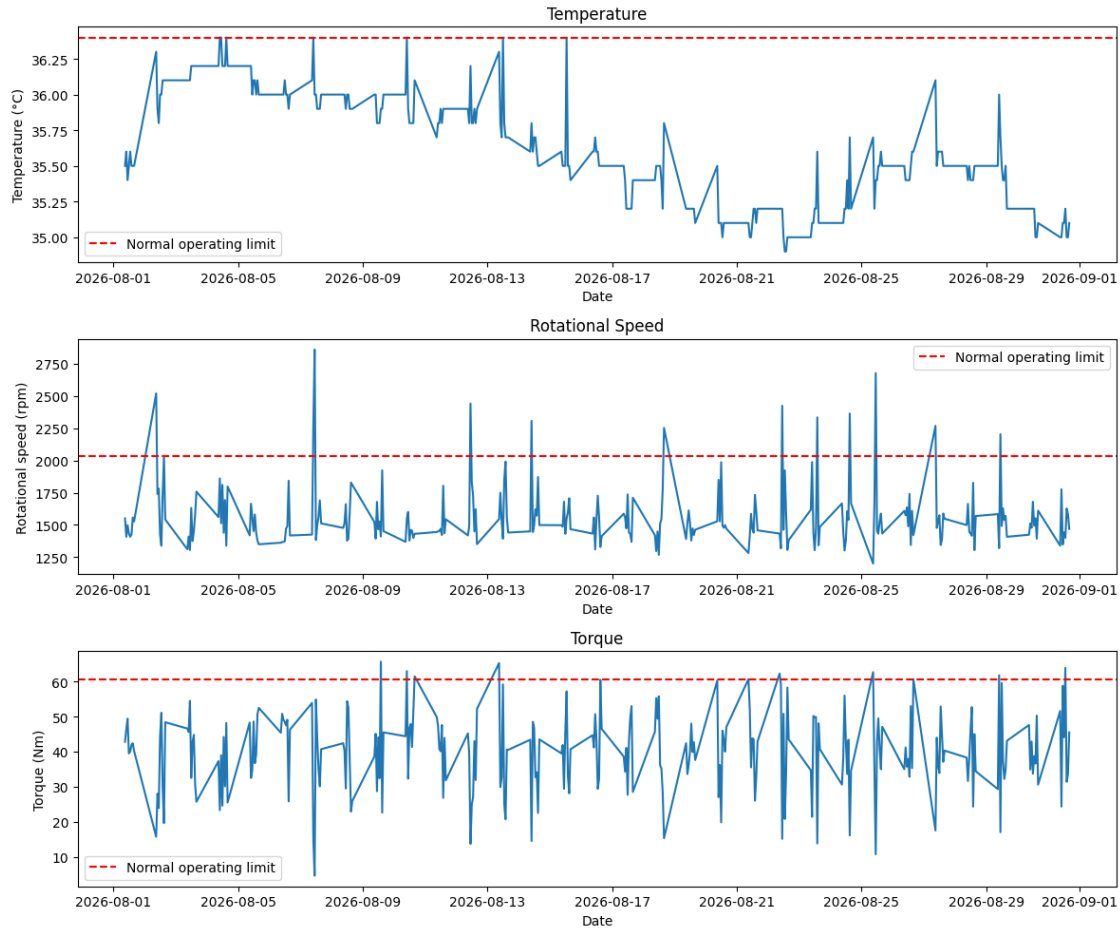
axes[0].set_ylabel("Temperature (°C)")
axes[0].legend()

# Rotational speed
axes[1].plot(Time, R_speed)
axes[1].set_title("Rotational Speed")
axes[1].axhline(
    limit_speed,
    linestyle="--",
    color="red",
    label="Normal operating limit"
)
axes[1].set_xlabel("Date")
axes[1].set_ylabel("Rotational speed (rpm)")
axes[1].legend()

# Torque
axes[2].plot(Time, torque)
axes[2].set_title("Torque")
axes[2].axhline(
    limit_torque,
    linestyle="--",
    color="red",
    label="Normal operating limit"
)
axes[2].set_xlabel("Date")
axes[2].set_ylabel("Torque (Nm)")
axes[2].legend()

plt.tight_layout()
plt.show()

```

1.9 2.9 Rotational Speed and Torque Comparison

The rotational speed and torque are plotted on the same graph to compare their evolution over time.

Because rotational speed and torque have different numerical scales, the torque values are multiplied by 30 for visualization purposes.

This scaling is used only for visualization and does not modify the original torque measurements.

```
[21]: plt.figure(figsize=(12, 5))

plt.plot(
    Time,
    R_speed,
    label="Rotational speed"
)

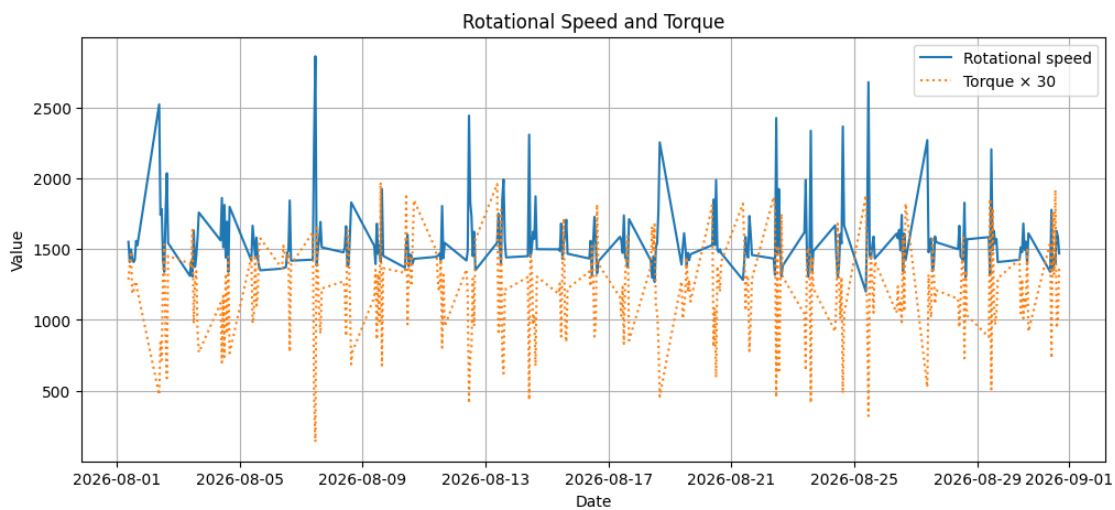
plt.plot(
```

```

    Time,
    torque * 30,
    linestyle=":",
    label="Torque × 30"
)

plt.xlabel("Date")
plt.ylabel("Value")
plt.title("Rotational Speed and Torque")
plt.legend()
plt.grid()
plt.show()

```



1.9.1 Interpretation

The comparison between rotational speed and torque shows that both parameters vary significantly during the analyzed period.

The rotational speed generally remains around 1,400–1,600 rpm, with several peaks exceeding 2,000 rpm. These peaks indicate periods of increased rotational speed that may require further investigation.

The torque also presents significant variations throughout the month. Since the torque values are multiplied by 30 in the graph, this scaling is used only to facilitate visual comparison and does not represent the actual torque values.

The two parameters do not appear to have a strong direct relationship in this visualization. Several high rotational-speed values occur when the torque is relatively low, while some higher torque values occur at moderate rotational speeds.

Therefore, rotational speed and torque should be analyzed together with other machine parameters, such as temperature and failure status, to better identify abnormal operating conditions.

[]: